Exercises

Exercise 4.1 · Quadratics. Solve in $\mathbb{C}$: $z^2+9=0$; $z^2-4z+13=0$; $2z^2+2z+1=0$; $z^4-1=0$.

Exercise 4.2 · Reducing to a quadratic.

1. Solve in $\mathbb{C}$ the equation $z^4+4z^2-5=0$ by setting $Z=z^2$.
2. Solve in $\mathbb{C}$ the equation $z+\dfrac{1}{z}=1$, for $z\neq 0$.

Exercise 4.3 · An obvious root. Let $P(z)=z^3-3z^2+4z-2$.

1. Check that 1 is a root of P.
2. Determine the polynomial Q such that $P(z)=(z-1)Q(z)$.
3. Solve $P(z)=0$ in $\mathbb{C}$.

Exercise 4.4 · A non-real root. Let $P(z)=z^3+z^2+4z+4$.

1. Check that $2i$ is a root of P. Deduce another root without computation.
2. Expand $(z-2i)(z+2i)$, then factorise $P(z)$ and solve $P(z)=0$.

Exercise 4.5 · Factorising z^3-27.

1. Factorise z^3-27 by $z-3$ using the course.
2. Solve in $\mathbb{C}$ the equation $z^3=27$ and plot the solutions in the complex plane. What do you notice?

Exercise 4.6 · A parameter. Let m be a real number. Consider the equation (E_m): $z^2-2z+m=0$.

1. Depending on the value of m, give the number and the nature of the solutions of (E_m).
2. Assume $m>1$. Let A and B be the images of the solutions. Determine m so that the triangle OAB is equilateral.

Exercise 4.7 · Recovering a polynomial. Determine the polynomial P of degree 3, with real coefficients and leading coefficient 1, such that $1+i$ is a root of P and $P(0)=4$.